

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Begins a proof using a valid method Eg. Factor theorem, algebraic division, multiplication of correct factors	AO1.1a	M1	$p\left(-\frac{1}{2}\right) = 30 \times \left(-\frac{1}{2}\right)^3 - 7\left(-\frac{1}{2}\right)^2 - 7\left(-\frac{1}{2}\right) + 2$ $= 0$ $\therefore 2x + 1 \text{ is a factor of } p(x)$
	Constructs rigorous mathematical proof. To achieve this mark: Factor theorem the student must clearly substitute and state that $p(-1/2) = 0$ and clearly state that this implies that $2x + 1$ is a factor Algebraic division OR Multiplication of correct factors The method must be completely correct with a concluding statement	AO2.1	R1	
(b)	Obtains quadratic factor PI	AO1.1a	M1	$p(x) = (2x + 1)(15x^2 - 11x + 2)$ $= (2x + 1)(5x - 2)(3x - 1)$
	Obtains second linear factor	AO1.1b	A1	
	Writes $p(x)$ as the product of the correct three linear factors. NMS correct answer 3/3	AO1.1b	A1	
(c)	Rearranges to achieve a cubic equation in $\sec x$ (or $\cos x$)	AO3.1a	M1	$\frac{30\sec^2 x + 2\cos x}{7} = \sec x + 1$ $\Rightarrow 30\sec^2 x + 2\cos x = 7\sec x + 7$ $\Rightarrow 30\sec^3 x + 2 = 7\sec^2 x + 7\sec x$ $30\sec^3 x - 7\sec^2 x - 7\sec x + 2 = 0$ $\Rightarrow (2\sec x + 1)(5\sec x - 2)(3\sec x - 1) = 0$ $\Rightarrow \sec x = -\frac{1}{2}, \frac{2}{3}, \frac{1}{5}$ These values do not fall within the range of $\sec x$ as they are
	Equates to zero and uses result from (b) or factorises	AO1.1a	M1	
	Deduces that if solutions exist they must be of the form $\sec x = -1/2$, $\sec x = 1/3$ or $\sec x = 2/5$ OE	AO2.2a	A1	
	Explains that the range of $\sec x$ is $(-\infty, -1] \cup [1, \infty)$ OE OE for $\cos x$	AO2.4	E1	
	Completes argument explaining that there cannot be any real solutions as values are	AO2.1	R1	

outside of the function's range			between -1 and 1 $\therefore \frac{30\sec^2 x + 2\cos x}{7} = \sec x + 1$ has no real solutions.
Total 10 marks			

Q2.

Marking Instructions	AO	Marks	Typical Solution
Recalls a correct trig identity, which could lead to a correct answer	AO1.2	B1	(LHS $\equiv$) $\cot^2 \theta - \cos^2 \theta$ $\equiv \frac{\cos^2 \theta}{\sin^2 \theta} - \cos^2 \theta$ $\equiv \cos^2 \theta \left(\frac{1}{\sin^2 \theta} - 1 \right)$ $\equiv \cos^2 \theta (\operatorname{cosec}^2 \theta - 1)$ $\equiv \cos^2 \theta \cot^2 \theta$ ($\equiv$ RHS) AG
Performs some correct algebraic manipulation and uses second identity to commence proof (at least two lines of argument)	AO2.1	R1	
Concludes a rigorous mathematical argument to prove given identity AG Must start with one side and through clear logical steps arrive at the other side. In order to be sufficiently clear, each line should be a single step, unless clear further explanation is given.	AO2.1	R1	
			Total 3 marks

Q3.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Uses the product rule for either term	AO1.1a	M1	$y = 2x\cos 3x + (3x^2 - 4)\sin 3x$ $\frac{dy}{dx} = 2\cos 3x - 6x\sin 3x$ $+ 6x\sin 3x + 3(3x^2 - 4)\cos 3x$ $= (9x^2 - 10)\cos 3x$
	Uses the product rule for both terms	AO1.1a	M1	
	Differentiates both terms correctly	AO1.1b	A1	
	Rearranges to correct form	AO1.1b	A1	

CAO			
(b) Finds $\frac{d^2y}{dx^2}$ from 'their' first derivative and equates to zero	AO3.1a	M1	$\frac{d^2y}{dx^2} = 18x\cos 3x - 3(9x^2 - 10)\sin 3x$
Applies product rule correctly on $\frac{dy}{dx}$ 'their' $\frac{dy}{dx}$ FT only applies if both M1 marks awarded in part (a)	AO1.1b	A1F	point of inflection $\Rightarrow \frac{d^2y}{dx^2} = 0 \Rightarrow$ $18x\cos 3x - 3(9x^2 - 10)\sin 3x = 0$ $\Rightarrow \frac{\cos 3x}{\sin 3x} = \frac{3(9x^2 - 10)}{18x}$
Arrives at a result using 'their' second derivative through correct algebraic manipulation that is correct for 'their' second derivative FT only applies if both previous marks in (b) have been awarded.	AO1.1b	A1F	$\Rightarrow \cot 3x = \frac{9x^2 - 10}{6x}$ (AG)
Constructs a clearly explained rigorous mathematical argument, to show the required result This must include a concluding statement or an explanation of reasoning at the start. AG	AO2.1	R1	
Total 8 marks			

Q4.

(a) $\left(\frac{dy}{dx}\right) = 3e^{3x} + \frac{1}{x}$

B1 for one term correct

B1

B1 all correct

B1

2

(b) (i) $\left(\frac{du}{dx}\right) = \frac{\pm \cos x(1 + \cos x) \pm \sin x(\sin x)}{(1 + \cos x)^2}$

*clear attempt at quotient / product rule
condone poor use of brackets*

M1

$$\frac{\cos x(1 + \cos x) - \sin x(-\sin x)}{(1 + \cos x)^2}$$

any correct form seen

A1

$$= \frac{\cos x + \cos^2 x + \sin^2 x}{(1 + \cos x)^2} = \frac{\cos x + 1}{(1 + \cos x)^2} = \frac{1}{1 + \cos x}$$

AG be convinced

correct use of brackets and correct notation used
throughout (eg A0 if $\cos x^2$ etc seen)

A1cso

3

(ii) $\left(\frac{dy}{dx}\right) \frac{1 + \cos x}{\sin x} \times \frac{1}{1 + \cos x}$ OE
correct use of chain rule

M1

$$= \frac{1}{\sin x}$$

$$= \operatorname{cosec} x$$

AG, must see = $\frac{1}{\sin x}$ and no errors seen;
condone incorrect use of brackets only if penalised in part
(b)(i)

A1

2

[7]

Q5.

(a) $\frac{\sec^2 x}{(\sec x + 1)(\sec x - 1)} = \frac{\sec^2 x}{\sec^2 x - 1}$

$\sec^2 x = 1 + \tan^2 x$ used

M1 for correct use of $\sec^2 x = 1 + \tan^2 x$ at least once or
($\operatorname{cosec}^2 x = 1 + \cot^2 x$)

M1

$$= \frac{\sec^2 x}{\tan^2 x} \text{ or } \frac{1 + \tan^2 x}{\tan^2 x}$$

$$\left(= \frac{1}{\cos^2 x \tan^2 x} \right)$$

$$= \frac{1}{\sin^2 x} \text{ or } \cot^2 x + 1$$

Shown, with no errors

A1

$$= \operatorname{cosec}^2 x$$

AG (No errors, omissions or poor notations seen)

A1

3

(b) $\operatorname{cosec}^2 x = \operatorname{cosec} x + 3$
 $\operatorname{cosec}^2 x - \operatorname{cosec} x - 3 = 0$

must have = 0

*correct solution of the quadratic,
or by completing the square*

B1

$$\operatorname{cosec} x = \frac{1 \pm \sqrt{13}}{2} \text{ or } (2.3... \text{ and } -1.3...)$$

$$\left(\operatorname{cosec} x = \pm \sqrt{\frac{13}{4} + \frac{1}{2}} \right)$$

PI by values for sin x

M1

$$\sin x = \frac{2}{1 \pm \sqrt{13}}$$

B1F for $\operatorname{cosec} x = \frac{1}{\sin^2 x}$ seen or implied

B1F

$$= 0.434 \text{ and } -0.768 \text{ (or } -0.767)$$

PI

A1

B1 for any three values correct AWRT

B1

$$x = 26^\circ, 154^\circ, -50^\circ, -130^\circ$$

*B1 for all four values correct AWRT and no extras
in the interval $-180 < x < 180^\circ$*

B1

6

(c) $2\theta - 60^\circ = x$

where x is a written value from candidate's (b) in degrees
PI by their answer

M1

$\theta = 43^\circ, 5^\circ$

CSO

A1

Ignore solutions outside interval $0^\circ < \theta < 90^\circ$

2

[11]

Q6.

(a) (i) $R = \sqrt{13}$

Accept 3.6 or better

B1

$\tan \alpha = \frac{2}{3}$
OE

M1

$\alpha = 33.7^\circ$

A1

3

(ii) minimum value = $-\sqrt{13}$

Accept -3.6 or better; ft R

B1ft

when $x - \alpha = \cos^{-1}(-1)$

NMS 0 / 2

M1

$x = 213.7^\circ$

Calculus used 0 / 2

A1

3

(b) (i) LHS = $\frac{\cos x}{\sin x} - 2 \sin x \cos x$

Express $\cot x - \sin 2x$ in terms of $\sin x$ and $\cos x$; ACF

M1

$$= \frac{\cos x}{\sin x} (1 - 2 \sin^2 x)$$

Factor out $\frac{\cos x}{\sin x}$ and $1 - 2 \sin^2 x$

m1

$$= \cot x \cos 2x$$

All correct

A1

3

Alternatives:

$$\text{RHS} = \cot x \cos 2x$$

$$= \frac{\cos x}{\sin x} (1 - 2 \sin^2 x)$$

*Express $\cot x \cos 2x$ in terms of $\cos x$ and $\sin x$,
 $\cos 2x$ ACF*

(M1)

$$= \frac{\cos x}{\sin x} - 2 \sin x \cos x$$

$\cos 2x = 1 - 2 \sin^2 x$ and multiply out and simplify.

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$$= \cot x - \sin 2x$$

All correct.

(A1)

(3)

$$\cot x (1 - \cos 2x) - \sin 2x = 0$$

$$\frac{\cos x}{\sin x} (1 - (1 - 2 \sin^2 x)) - 2 \sin x \cos x = 0$$

Rearrange to expression = 0 and factor out $\cot x$;

*Express $\cot x$, $\cos 2x$ and $\sin 2x$ in terms of $\sin x$ and
 $\cos x$, ACF*

(M1)

$$\frac{\cos x}{\sin x} (2\sin^2 x) - 2\sin x \cos x = 0$$

$$\cos 2x = 1 - 2\sin^2 x \text{ used}$$

(m1)

$$2\sin x \cos x - 2\sin x \cos x = 0$$

Simplified, with all correct

(A1)

(3)

(ii) $\cot x - \sin 2x = 0$

$$\cot x \cos 2x = 0$$

$$\cot x = 0 \quad \text{or} \quad \cos 2x = 0$$

Both equations correct

M1

$$2x = 90^\circ \text{ (270}^\circ\text{)}$$

Condone missing 270°

m1

$$x = 90^\circ, 45^\circ, 135^\circ$$

All correct

A1

3

Alternatives:

$$\cot x - \sin 2x = \frac{\cos x}{\sin x} - 2\sin x \cos x = 0$$

$$\cos x \left(\frac{1}{\sin x} - 2\sin x \right) = 0$$

$$\cos x = 0 \quad \text{or} \quad 1 - 2\sin^2 x = 0$$

Both equations

(M1)

$$\sin x = (\pm) \frac{1}{\sqrt{2}}$$

(m1)

$$x = 90^\circ, 45^\circ, 135^\circ$$

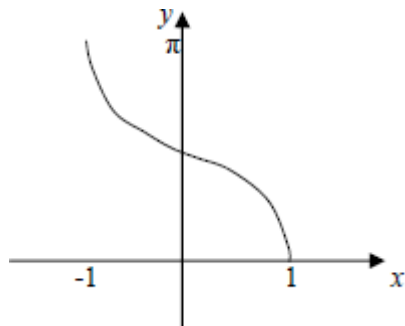
(A1)

(3)

[12]

Q7.

(a)



Correct sketch of $\cos^{-1} x$.

B1

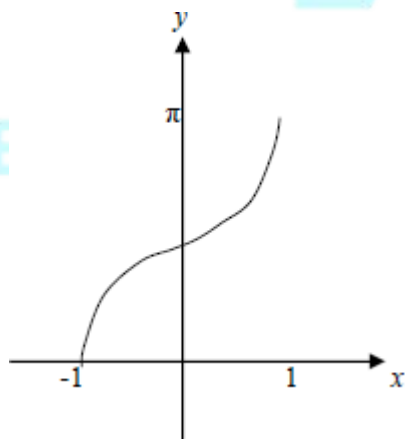
$(-1, \pi)$ and $(1, 0)$

Stated

B1

2

(b)



*Correct sketch of $\pi - \cos^{-1} x$
Must touch negative x -axis.*

B1

$(-1, 0)$ and $(1, \pi)$

Stated

B1

2

Q8.

$$(a) \text{ LHS} = \frac{(1 - \cos \theta) + (1 + \cos \theta)}{(1 + \cos \theta)(1 - \cos \theta)}$$

Combining fractions

M1

$$= \frac{2}{1 - \cos^2 \theta}$$

Correctly simplified

A1

$$= \frac{2}{\sin^2 \theta}$$

Use of $\sin^2 \theta + \cos^2 \theta = 1$

m1

$$2 \operatorname{cosec}^2 \theta = 32$$

$$\operatorname{cosec}^2 \theta = 16$$

AG; no errors seen

A1

OR

$$1 - \cos \theta + 1 + \cos \theta = 32(1 + \cos \theta)(1 - \cos \theta) \text{ (M1)}$$

$$2 = 32(1 - \cos^2 \theta) \text{ (A1)}$$

$$2 = 32 \sin^2 \theta \text{ (m1)}$$

$$\operatorname{cosec}^2 \theta = 16 \text{ (A1)}$$

$$(b) \operatorname{cosec} y = (\pm) \sqrt{16} \text{ or better (PI by further working)}$$

$$\text{or } \sin y = (\pm) \sqrt{\frac{1}{16}} \text{ or better}$$

M1

$$(y =)$$

$$0.253, (2.889,) (3.394,) (6.031,) (-0.253)$$

Sight of any of these correct to 3dp or better

B1

$$(y =)$$

$$0.25, 2.89, 3.39 \text{ (or better)}$$

Must see these 3 answers, with or without either / both of -0.25 or 6.03

Ignore answers outside interval -0.25 to 6.03 but

extras in this interval scores A0

A1

$$x = 0.43, 1.74, 2(.00), 0.17$$

3 correct (must be 2 dp)

All 4 correct (must be 2 dp) and no extras in interval
(ignore answers outside interval)

B1
B1

5

[9]

Q9.

(a) $\left(\frac{dx}{dy} = \right) \frac{\cos y \times \cos y - \sin y \times -\sin y}{\cos^2 y}$

Condone incorrect signs, poor notation, omission of
 $\frac{dy}{dx}$ or using $\frac{dx}{dy}$

M1

$$= \frac{\cos^2 y + \sin^2 y}{\cos^2 y}$$

RHS correct with terms squared, including correct notation
Must see this line

A1

$$= \frac{1}{\cos^2 y} \text{ or } (= 1 + \tan^2 y)$$

Must see one of these

$$\frac{dx}{dy} = \sec^2 y$$

AG; all correct including correct use of $\frac{dx}{dy}$ throughout

A1
CSO

3

(b) $\sec^2 y = 1 + (x - 1)^2$

Correct use of $\sec^2 y = 1 + \tan^2 y$ and in terms of x

M1

$$= 1 + x^2 - 2x + 1 \quad \text{OE}$$

$$= x^2 - 2x + 2$$

AG; must see " $\sec^2 y =$ ", $(x - 1)^2$ expanded and no errors seen

A1

2

(c) $\frac{dx}{dy} = x^2 + 2x + 2$ or $\frac{dy}{dx} = \frac{1}{\sec^2 y}$

Must be seen

$$\frac{dy}{dx} = \frac{1}{x^2 - 2x + 2}$$

AG and no errors seen

B1

1

(d) (i) $y = \tan^{-1}(x - 1) - \ln x$

$$\left(\frac{dy}{dx} \right) = \frac{1}{x^2 - 2x + 2} - \frac{1}{x}$$

Must be correct

M1

$$\left(\frac{dy}{dx} = 0 \right)$$

$$\pm x^2 + bx + c (= 0)$$

Expression in this form (generous), where b and $c \neq 0$

m1

$$x^2 - 3x + 2 = 0$$

Must see correct equation = 0

A1

$$x = 1, 2$$

Both answers must be seen

The two A marks are independent

A1

4

(ii) $y'' = p(x^2 - 2x + 2)^{-2}(2x - 2) \pm qx^{-2}$
where p and q are constants

M1

$$y'' = -(x^2 - 2x + 2)^{-2} (2x - 2) + x^{-2}$$

$p = -1, q = 1$ including correct brackets

A1

2

(iii) $x = 1, y'' = 1$

Must have scored full marks in (d)(i) and (ii)

M1

At $x = 1, y'' > 0 \therefore$ min

When $x = 1, y = 0$ hence on x -axis

*Must see $y'' > 0$ or in words
Both statements fully correct*

A1

2

[14]

Q10.

(a) (i) $\frac{dx}{dy} = k \sec^2(3y + 1)$

Where k is an integer

*Condone omission of $\frac{dx}{dy}$
But*

$$\frac{dy}{dx} = k \sec^2(3y + 1) \quad \text{scores}$$

M1 A0

M1

$$= 3 \sec^2(3y + 1)$$

ISW

A1

Alternative methods

$$y = \frac{1}{3}(\tan^{-1}x - 1)$$

$$\frac{dx}{dy} = k(1 + x^2)$$

M1

$$3 = (1 + \tan^2(3y + 1))$$

A1

Or

$$x = \frac{\sin(3y + 1)}{\cos(3y + 1)}$$

$$\frac{dx}{dy} = \frac{\pm k \cos^2(3y+1) \pm k \sin^2(3y+1)}{\cos^2(3y+1)} \quad \text{M1}$$

$$= \frac{3}{\cos^2(3y+1)} \quad \text{A1}$$

2

(ii) $\frac{dx}{dy} = 3 \sec^2\left(3x - \frac{1}{3} + 1\right) = 3 \sec^2 0$

Substitution of $y = -\frac{1}{3}$ into their $\frac{dx}{dy}$ **or**
 $\frac{dy}{dx}$

BUT must have scored M1 in (a)(i)

M1

$$\frac{dy}{dx} = \frac{1}{3} \quad \text{CSO}$$

Condone 0.333 or better

Or

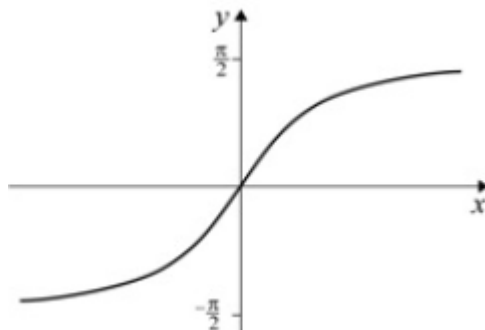
$$\left. \begin{aligned} \frac{dy}{dx} &= \frac{1}{3 \sec^2(3y+1)} \\ &= \frac{1}{3 \sec^2 0} \\ &= \frac{1}{3} \end{aligned} \right\} \text{As above}$$

A1

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2

(b)



*Approx correct shape with no turning points,
 through (0,0) and only 1 curve*

Asymptotic at both $\pm \frac{\pi}{2}$ and both values shown

M1

Condone ± 90 (degrees)

Condone $y = \tan x$ also drawn but clearly identified, otherwise M0

A1

2

[6]

Q11.

(a) $\cos x = -0.2$

Or $\tan x = (\pm) \sqrt{24}$

M1

$x = 1.77, 4.51$

AWRT

One correct value

A1

Second correct value and no extra values in interval 0 to 6.28...

Ignore answers outside interval

SC

$x = 1.8, 4.5$ with or without working M1 A1 A0

SC (using degrees)

101.54, 281.54

M1 A1 A0

101.5, 281.5

M1 A0 A0

SC

No working shown

2 correct answers 3/3

1 correct answer 2/3

A1

3

(b) LHS

$$= \frac{\operatorname{cosec} x (1 - \operatorname{cosec} x) - \operatorname{cosec} x (1 + \operatorname{cosec} x)}{(1 + \operatorname{cosec} x)(1 - \operatorname{cosec} x)}$$

Correctly combining fractions but condone poor use, or omission, of brackets

M1

$$= \frac{\operatorname{cosec} x - \operatorname{cosec}^2 x - \operatorname{cosec} x - \operatorname{cosec}^2 x}{1 - \operatorname{cosec}^2 x}$$

Allow recovery from incorrect brackets

A1

$$= \frac{-2 \operatorname{cosec}^2 x}{-\cot^2 x} \text{ or } \frac{-2(1 + \cot^2 x)}{-\cot^2 x}$$

Correct use of relevant trig identity

eg $\operatorname{cosec}^2 x = 1 + \cot^2 x$

m1

$$2 \sec^2 x = 50$$

$$\sec^2 x = 25$$

AG

*All correct with no errors seen INCLUDING
correct brackets on 1st line*

A1

4

Or

$$\frac{\operatorname{cosec} x}{1 + \operatorname{cosec} x} - \frac{\operatorname{cosec} x}{1 - \operatorname{cosec} x} = 50$$

$$\begin{aligned} \operatorname{cosec} x (1 - \operatorname{cosec} x) - \operatorname{cosec} x (1 + \operatorname{cosec} x) \\ = 50 (1 + \operatorname{cosec} x)(1 - \operatorname{cosec} x) \end{aligned}$$

*Correctly eliminating fractions but
condone poor use, or omission, of brackets*

(M1)

$$\begin{aligned} \operatorname{cosec} x - \operatorname{cosec}^2 x - \operatorname{cosec} x - \operatorname{cosec}^2 x \\ = 50 (1 - \operatorname{cosec}^2 x) \end{aligned}$$

Allow recovery from incorrect brackets

(A1)

$$48 \operatorname{cosec}^2 x = 50$$

$$\sin^2 x = \frac{24}{25} \Rightarrow \cos^2 x = \frac{1}{25}$$

Correct use of relevant trig identity

eg $\sin^2 x = 1 - \cos^2 x$

(m1)

$$\sec^2 x = 25$$

AG

All correct with no errors seen INCLUDING
correct brackets on 1st line

(A1)

(c) $\sec x = \pm 5$

Or $\cos x = \pm 0.2$

Or $\tan x = \pm \sqrt{24}$

M1

$x = 1.77, 4.51, 1.37, 4.91$

(AWRT)

3 correct

A1

4 correct and no other answers in interval
Ignore answers outside interval

SC

1.8, 4.5, 1.4, 4.9

With or without working

M1 A1

SC

their 2 answers from (a)

+1.37, 4.91

(AWRT)

2/3

SC For this part, if in degrees

max mark is

M1 A0

SC

No working shown

4 correct answers 3/3

3 correct answers 2/3

0, 1, 2 correct answers 0/3

A1

3

[10]

Q12.

(a) (i) $(\sin^{-1} \pm 0.25) \pm 14.5$

PI by sight of 194.5 etc

condone ± 14.4

M1

$\theta = 194.5, 345.5$ (AWRT)

no extras in interval, ignore answers outside interval

A1

2

(ii) $2 \cot^2 (2x + 30) = 2 - 7 \operatorname{cosec} (2x + 30)$

condone replacing $2x + 30$ by Y

$$2(\operatorname{cosec}^2 (2x + 30) - 1) = 2 - 7 \operatorname{cosec} (2x + 30)$$

correct use of $\operatorname{cosec}^2 Y = 1 + \cot^2 Y$

M1

$$2 \operatorname{cosec}^2 (2x + 30) + 7 \operatorname{cosec} (2x + 30) - 4 (= 0)$$

must be in this form

A1

$$(2 \operatorname{cosec} (2x + 30) \pm 1)(\operatorname{cosec} (2x + 30) \pm 4) (= 0)$$

attempt at factorisation

m1

$$\operatorname{cosec}(2x + 30) = \frac{1}{2} \text{ or } -4$$

must be this line using $f(2x + 30)$

A1

$$2x + 30 = 194.5, 345.5$$

$$x = 82.2, 157.8 \text{ (AWRT)}$$

one correct answer, allow 82.3, ignore extra solutions

B1

CAO both answers correct and no extras in interval, ignore answers outside interval

B1

6

(b) stretch (I)

scale factor $\frac{1}{2}$ (II)

parallel to x -axis (III)

I and either II or III

M1

I + II + III

A1

translate

E1

$$\begin{pmatrix} -15 \\ 0 \end{pmatrix}$$

condone '15 to left' or '-15 in x (direction)'

B1

4

alternative method

translate

(E1)

$$\begin{pmatrix} -30 \\ 0 \end{pmatrix}$$

stretch

as above

(B1)

(M1)

scale factor $\frac{1}{2}$

parallel to x -axis

as above

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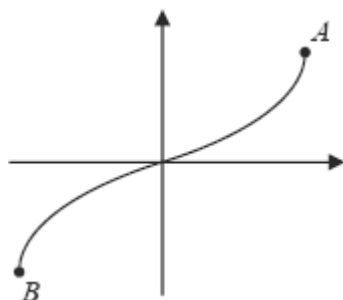
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(A1)

[12]

Q13.

(a) (i)



correct shape passing through origin

and stopping at A and B

B1

$$A\left(1, \frac{\pi}{2}\right)$$

B1

$$B\left(-1, -\frac{\pi}{2}\right)$$

SC $A(1, 90)$ and $B(-1, -90)$ scores B1

B1

3

- (ii) line intersecting their curve (positive gradient, positive y intercept)

M1

Correct statement

one solution only, stated or indicated on sketch - must be in the first quadrant (ie curve intersects line once)
Must have scored B1 for graph in (a)(i)

A1

2

(b)
$$\left. \begin{array}{ll} \text{LHS}(0.5) = 0.5 & \text{RHS}(0.5) = 1.1 \\ \text{LHS}(1) = 1.6 & \text{RHS}(1) = 1.3 \end{array} \right\}$$

M1

At 0.5 LHS < RHS, At 1 LHS > RHS

$$\therefore 0.5 < \alpha < 1$$

CSO

A1

or

$$f(x) = \sin^{-1}(x) - \frac{1}{4}x - 1$$

$f(x)$ must be defined

$$\left. \begin{array}{l} f(0.5) = -0.6 \\ f(1) = 0.3 \end{array} \right\} \text{AWRT}$$

Allow $f(0.5) < 0$ $f(1) > 0$

(M1)

Change of sign $\Rightarrow 0.5 < \alpha < 1$

(A1)

or

$$f(x) = \sin\left(\frac{1}{4}x + 1\right) - x$$

f(x) must be defined

$$\left. \begin{array}{l} f(0.5) = 0.4 \\ f(1) = -0.1 \end{array} \right\} \text{Attempt}$$

(M1)

Change of sign $\Rightarrow 0.5 < \alpha < 1$

(A1)

or

$$f(x) = 4 \sin^{-1} x - x - 4$$

f(x) must be defined

$$\left. \begin{array}{l} f(0.5) = -2.4 \\ f(1) = 1.3 \end{array} \right\} \text{Attempt}$$

(M1)

Change of sign $\Rightarrow 0.5 < \alpha < 1$

(A1)

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2

(c) (i) $x_2 = 0.902$

Sight of AWRT 0.902 or AWRT 0.941

M1

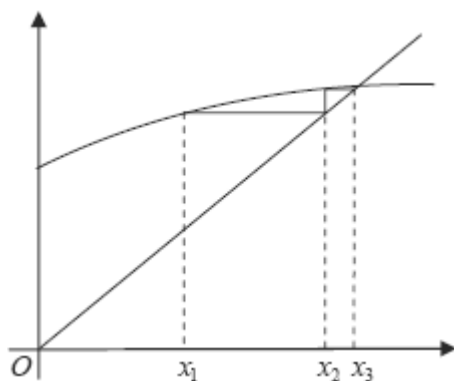
$$x_3 = 0.941$$

These values only

A1

2

(ii)



*Staircase, (vertical line) from x_1 to curve,
horizontal to line, vertical to curve*

M1

x_2, x_3 approx correct position on x -axis

A1

2

[11]

Q14.

(a) $(y =) 1$

Condone 1 marked at A, $A = 1$ etc

but not $\frac{1}{\cos 0}$, sec 0

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B1

1

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(b) $\cos x = \frac{1}{2}$ or $\cos^{-1} \frac{1}{2}$ seen

or sight of $\pm 60^\circ$ or $\pm \frac{\pi}{3}$, ± 1.05 (AWRT)

M1

$x = 60^\circ, 300^\circ$

Condone extra values outside

$0^\circ < x < 360^\circ$, but no extras in interval

A1

2

[3]

Q15.

(a) $\sec x = \frac{3}{2}$
 $\cos x = \frac{2}{3}$

$x = 48, 312$

1 correct

B1

(Condone answers rounding to)

2 correct and no extras in interval

B1

2

(b) $2 \tan^2 x = 10 - 5 \sec x$
 $2 (\sec^2 x - 1) = 10 - 5 \sec x$

Use of trig identity correctly

M1

$2 \sec^2 x + 5 \sec x - 12 (= 0)$

A1

$(2 \sec x - 3) (\sec x + 4) (= 0)$

*Attempt to solve or factorise
 1 slip using formula*

m1

$\left. \begin{array}{l} \sec x = \frac{3}{2}, -4 \\ \cos x = \frac{2}{3}, -\frac{1}{4} \end{array} \right\} \text{either of these}$

A1

$x = 48, 312, 104, 256$

AWRT 3 correct condone 105 or 255

B1

All correct and no extras in interval

B1

Alternative:

$$\left. \begin{aligned}
 \frac{2 \sin^2 x}{\cos^2 x} &= 10 - \frac{5}{\cos x} \\
 2 \sin^2 x &= 10 \cos^2 x - 5 \cos x \\
 2 - 2 \cos^2 x &= 10 \cos^2 x - 5 \cos x
 \end{aligned} \right\}$$

(M1)(A1)

$12 \cos^2 x - 5 \cos x - 2 = 0$
 then rest of scheme as above

6

[8]

Q16.

(a) $P(-1, \pi)$

Condone $(-1, 180^\circ)$

B1

$Q(1, 0)$

B1

2

(b) Translate

E1

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

or equivalent in words

B1

Stretch SF 2 // y -axis

Stretch + one other correct

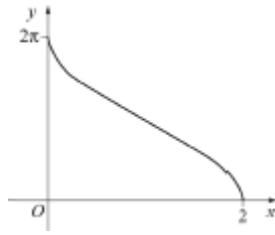
M1

all correct

A1

4

(c)



Correct shape in 1st quadrant

B1

2π and 2 marked correctly

B1

2

(d) (i) $\frac{y}{2} = \cos^{-1}(x - 1)$

M1

$$\cos\left(\frac{y}{2}\right) = x - 1$$

$$x = \cos\left(\frac{y}{2}\right) + 1$$

A1

2

(ii) $-\frac{1}{2} \sin\left(\frac{y}{2}\right)$

k sin (...)

M1

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$\frac{dx}{dy}$ correct

A1

$$\text{At } y = 2, \left(\frac{dx}{dy}\right) = -\frac{1}{2} \sin 1$$

Condone AWRT -0.42

A1

3

[13]

Q17.

(a) $\cot x = 2 \Rightarrow \tan x = 0.5$

M1

$$x = 0.46, 3.61$$

AWRT; no others within range

A1

2

(b) $\operatorname{cosec}^2 x = \frac{3 \cos x + 4}{2}$

$$2(1 + \cot^2 x) = 3 \cot x + 4$$

Correct use of $\operatorname{cosec}^2 x = 1 + \cot^2 x$

M1

$$(2 \cot^2 x - 3 \cot x + 2 - 4 = 0)$$

$$2 \cot^2 x - 3 \cot x - 2 = 0$$

*AG; correct with no slips from line
with no fractions*

A1

2

(c) $(2 \cot x + 1)(\cot x - 2) (= 0)$

Attempt to solve

M1

$$\cot x = -\frac{1}{2}, 2$$

A1

$$\tan x = -2, 0.5$$

$$x = 0.46, 3.61, 2.03, 5.18$$

2 correct Allow 3.6(0)

B1

4 correct (with no extras in range) AWRT

SC Degrees

$$\left. \begin{array}{l} 26.57, 206.57 \\ 116.57, 296.57 \end{array} \right\} \text{B1 For 2 correct}$$

B1

4

[8]

Q18.

(a) $x = \cos^{-1} \frac{1}{3}$
PI

M1

$= 1.23, 5.05 (0.39\pi, 1.61\pi)$
AWRT (-1 for each error in range)
SC 70.53, 289.47 B1

A1,A1

3

(b) $\sec^2 x - 1 = 2 \sec x + 2$
use of $\sec^2 x = 1 + \tan^2 x$

M1

$\sec^2 x - 2 \sec x - 3 = 0$
AG; CSO

A1

2

(c) $\sec^2 x - 2 \sec x - 3 = 0$
 $(\sec x - 3)(\sec x + 1) = 0$
attempt to solve

M1

$\cos x = \frac{1}{3} \text{ or } -1$ o.e

A1

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$x = 1.23, 5.05,$
(2 answers in range from (a)) AWRT

B1F

$3.14 (\pi)$
all correct and no extras in range
SC 70.53, 289.47, 180 B1

B1

4

[9]

Q19.

Stretch (I)

SF $\frac{1}{3}$ (II)

For I + (II or III)

M1

Parallel to x -axis (III)

All correct

A1

Translate

Allow translation

E1

$\begin{pmatrix} 0 \\ 1 \end{pmatrix}$

Correct vector or description

B1

4

[4]

Q20.

(a) $A(-1, \pi)$

B1

$B\left(0, \frac{\pi}{2}\right)$

B1

2

(b) $\cos^{-1} x - 3x - 1 = 0$

$f(0.1) = 0.17$ allow 0.2, 0.1

Or comparing 'sides'

M1

$f(0.2) = -0.23$ allow -0.2

Change of sign $\therefore$ root

A1

2

(c) $x_1 = 0.1$

M1

$$x_2 = 0.1569 = 0.157$$

A1

$$x_3 = 0.1378 = 0.138$$

$$x_4 = 0.144$$

A1

3

[7]

Q21.

(a) $\operatorname{cosec} x = 2$

$$\Rightarrow \sin x = \frac{1}{2}$$

30° scores M1 implied

M1

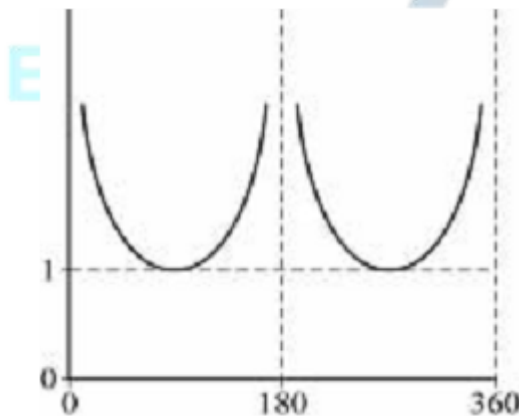
$$= 30, 150$$

and no extras in range

A1

2

(b)



all positive, 2 U shapes

M1

*minima consistent > 0 , **not** intersecting with each other or y-axis*

A1

2

[4]

Q22.

(a) $\left(1, \frac{\pi}{2}\right)$ *OE in decimals*

Or for -1 and 1

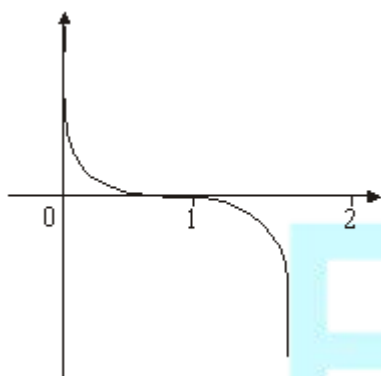
B1

$\left(-1, -\frac{\pi}{2}\right)$

B1

2

(b)



Translation in +ve x direction

M1

Correct shape

M1

Correct Graph

Through (1,0) touching y-axis

A1

3

[5]

Q23.

(a) $\sec x = 5$
 $\cos x = 0.2$

M1

$x = 1.37, 4.91$ AWRT

A1A1

3

(b) $\tan^2 x = 3 \sec x + 9$
 $\sec^2 x - 1 = 3 \sec x + 9$

for using $\sec^2 x = 1 + \tan^2 x$ OE

M1

$$\sec^2 x - 3 \sec x - 10 = 0$$

AG

A1

2

(c) $(\sec x - 5)(\sec x + 2) = 0$
or use of formula (attempt)

M1

$$\sec x = 5, -2$$

A1

$$\cos x = 0.2, -0.5$$

$$x = 1.37, 4.91$$

any 2 correct or ft their 2 answers in (a)

B1F

$$2.09, 4.19$$

all 4 correct, no extras

A1

4

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[9]

Q24.

(a) $\tan^2 x = \sec x + 11$

Or attempt to form quadratic in $\cos^2$

$$\sec^2 x - 1 = \sec x + 11$$

$$\tan^2 x = \sec^2 x - 1$$

M1

$$\sec^2 x - \sec x - 12 = 0$$

AG

A1

2

- (b) $(\sec x - 4)(\sec x + 3) = 0$
attempt at solving quadratic

M1

$$\sec x = 4, -3$$

A1F

$$\therefore \cos x = \frac{1}{4}, -\frac{1}{3}$$

$$AG; (A0 \text{ if no use of } \cos x = \frac{1}{\sec x})$$

A1

3

- (c) $x = 76^\circ, 284^\circ$
2 correct

B1

$$x = 109^\circ, 251^\circ \text{ (or better)}$$

other answers (−1 each extra in range)

B1,B1

3

If radians $x = 1.32, 4.97$
 $1.91, 4.37$
B1 any 2 correct
B1 other 2 correct

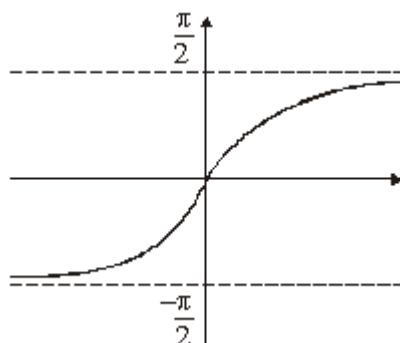
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[8]

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Q25.

(a)



shape

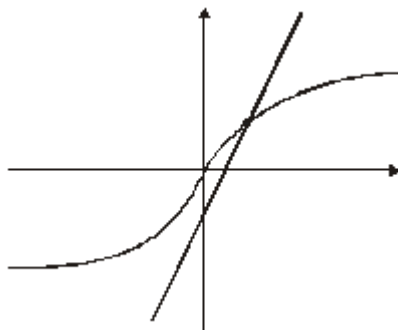
B1

asymptotes (shown or stated) ($\frac{\pi}{2}$ seen)

B1

2

(b) (i)



sketch of $2x - 1$

correct

B1

B1

2

(ii) $\tan^{-1} x - 2x + 1 = 0$

$f(0.8) = 0.07$

M1

$f(0.9) = -0.07$

change of sign $\therefore$ root

allow +ve, -ve

A0 if $f(0.8)$, $f(0.9)$ wrong

A1

2

(c) ($x_1 = 0.8$)

attempt at x_2

M1

$x_2 = 0.837$ (37) ...

for x_2

A1

$x_3 = 0.85$

for x_3

A1

3

[9]



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